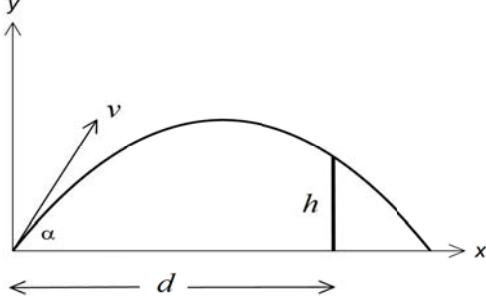


Question No	Question	Mark
1	a) Differentiate $(x^2 + 2x + 2)e^{-x}$	3
	b) Hence evaluate $\int_1^2 x^2 e^{-x} dx$	2
2	Find $\int \frac{\cos x}{1 + \sin^2 x} dx$ by using the substitution $u = \sin x$	3
3	a) Let $P(\theta) = 2 \sin \theta - \theta$. Show that $P(\theta)$ has a root between 1 and 2.	2
	b) Taking $\theta = 1.8$ as an initial approximation for the solution to the equation $2 \sin \theta = \theta$ between $\frac{\pi}{2}$ and π , use one application of Newton's Method to find a closer approximation.	2
4	A spherical balloon is being inflated so that its surface area is increasing at a steady rate of $12\text{cm}^2/\text{s}$.	
	a) Show that $\frac{dr}{dt} = \frac{3}{2\pi r}$ where r is the radius of the balloon.	2
	b) Find the rate of increase of the volume at the instant that the surface area is $1600\pi\text{cm}^2$.	3
5	(i) Prove that $\frac{d^2x}{dt^2} = \frac{d(\frac{1}{2}v^2)}{dx}$ where v and x are the velocity and displacement of a particle respectively.	2
	(ii) A particle moves in a straight line with an acceleration given by $\ddot{x} = -\frac{1}{2}e^{-x}$. If $v = 1$ and $x = 0$ when $t = 0$, find the	
	a) velocity in terms of x	2
	b) displacement in terms of t	3

6	The speed v m/s of a particle moving along the x axis is given by $v^2 = 18 + 32x - 8x^2$, where x is the distance from the origin. a) Prove that the particle is in Simple Harmonic Motion. b) Find its centre of motion and its period. c) What is the amplitude?	2 2 2
7	When the trigger of an aerosol can is depressed, the rate of decrease of the gas pressure in the can is given by $\frac{dp}{dt} = k\sqrt{p-1}$ where p is the pressure (in atmospheres) in the can after t seconds. Initially the pressure is 10 atmospheres and after 10 seconds the pressure is 5 atmospheres. a) Calculate the value of k. b) Find the pressure in the can after using it for 20 seconds.	3 2
8	A ball is projected at an angle of elevation α with speed V from a point on the ground.  i) The equations of motion are given as: $x = Vt \cos \alpha$ $y = Vt \sin \alpha - \frac{1}{2}gt^2.$ Prove that the maximum height reached is $\frac{V^2 \sin^2 \alpha}{2g}$ ii) The ball just clears a wall of height h m at a distance of d m from the initial point of projection. Prove that the maximum height reached by the ball is: $\frac{1}{4} \left(\frac{d^2 \tan^2 \alpha}{d \tan \alpha - h} \right)$	2 3
End of Task		

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln(x + \sqrt{x^2 - a^2}), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln(x + \sqrt{x^2 + a^2})$$

NOTE: $\ln x = \log_e x, \quad x > 0$



BAULKHAM HILLS HIGH SCHOOL

2013

HSC Assessment Task 3

Mathematics Extension 1

General Instructions

- Reading time – 5 minutes
- Working time – 60 minutes
- Write using black or blue pen
- Board-approved calculators may be used
- All necessary working should be shown in every question
- Marks may be deducted for careless or badly arranged work
- Attempt all questions

Total marks – 40

This paper consists of EIGHT questions on pages 3-4.

Attempt all questions

Answer each question on the appropriate page in the answer booklet

A page of standard integrals is on page 2.

Student's Number:.....

Teacher:.....TEACHER.



Year 12
HSC Assessment Task - 3
June 2013
MATHEMATICS EXTENSION 1

Marking Cover Sheet

Basic Skills		Problem Solving	
Question	Mark	Question	Mark
1 a)	/3	1 b)	/2
		2	/3
3 b)	/2	3 a)	/2
4 a)	/2	4 b)	/3
5 i)	/2	5 ii) a, b	/5
6 a)	/2	6 b), c)	/4
7 b)	/2	7 a)	/3
8 i)	/2	8 ii)	/3
SUB TOTAL	/15	SUB TOTAL	/25
TOTAL	/40		

Question 1

BOS#:

$$a) \frac{d}{dx} (x^2 + 2x + 2) e^{-x}$$

$$= e^{-x} [2x + 2] + (x^2 + 2x + 2)(-1) e^{-x}$$

$$= e^{-x} (2x + 2 - x^2 - 2x - 2)$$

$$= -x^2 e^{-x} \checkmark$$

(3)

$$b) \int_1^2 e^{-x} x^2 = \left[-(x^2 + 2x + 2) e^{-x} \right]_1^2$$

$$= -10 e^{-2} + 5 e^{-1}$$

$$= \frac{5}{e} - \frac{10}{e^2} \checkmark$$

(2)

$$= 1.839 - 1.3534$$

$$= 0.4856$$

You may ask for extra writing paper if you need more space to answer question 1

Question 2

BOS#:

$$u = \sin x$$

$$\frac{du}{dx} = \cos x \checkmark$$

or

$$\int \frac{\cos x}{1 + \sin^2 x} dx = \int \frac{du}{1 + u^2} \checkmark$$

(3)

$$= \tan^{-1}\left(\frac{u}{1}\right) + C \checkmark$$

$$= \tan^{-1}(\sin x) + C \checkmark \leftarrow \begin{array}{l} \text{Replace } u \\ \text{and} \\ \text{include } C. \end{array}$$

You may ask for extra writing paper if you need more space to answer question 2

Question 3

BOS#:

$$a) P(\theta) = 2 \sin \theta - \theta$$

$$P(1) = 2 \sin(1) - 1 = 0.6829$$

$$P(2) = 2 \sin(2) - 2 = -0.1814$$

$P(1)$ and $P(2)$ have opposite signs and $P(\theta)$ is continuous in $1 \leq \theta \leq 2$, $\exists$ a root between 1 and 2.

$$b) \theta_0 = 1.8$$

$$\theta_1 = \theta_0 - \frac{P(\theta)}{P'(\theta)}$$

$$= 1.8 - \frac{[2 \sin(1.8) - 1.8]}{[2 \cos(1.8) - 1]}$$

$$= 1.8 - \frac{0.1477}{-1.4544} \quad \text{or}$$

$$= 1.9 \quad (\text{old}) \checkmark$$

You may ask for extra writing paper if you need more space to answer question 3

Question 4

BOS#:

$$a) \quad A = 4\pi r^2 \quad V = \frac{4}{3}\pi r^3$$

$$\frac{dA}{dt} = 12 \text{ cm}^2/\text{s}$$

$$\frac{dA}{dt} = 4\pi \cdot 2r \cdot \frac{dr}{dt} \Rightarrow \frac{3}{2} = 4\pi \cdot 2r \cdot \frac{dr}{dt}$$

$$\frac{dr}{dt} = \frac{3}{2\pi r}$$

$$b) \quad \frac{dv}{dt} = \frac{4}{3}\pi \cdot 2r^2 \cdot \frac{dr}{dt}$$

$$= 4\pi r^2 \cdot \frac{dr}{dt} \checkmark$$

$$\text{when } A = 1600\pi \text{ m}^2$$

$$1600\pi = 4\pi r^2$$

$$r = 20 \checkmark$$

$$\therefore \frac{dv}{dt} = \frac{1600\pi \cdot 3}{2\pi \times 20} = 120$$

$$\therefore \text{rate of increase} = 120 \text{ cm}^3/\text{s} \checkmark$$

You may ask for extra writing paper if you need more space to answer question 4

Question 5

BOS#:

$$(i) \quad \frac{d^2x}{dt^2} = \frac{d(\frac{1}{2}v^2)}{dx}$$

$$\text{LHS} = \frac{dv}{dt} \checkmark$$

$$= \frac{dv}{dx} \cdot \frac{dx}{dt}$$

$$= v \cdot \frac{dv}{dx} \checkmark \quad \text{where } \frac{dx}{dt} = v$$

$$= \frac{d(\frac{1}{2}v^2)}{dx}$$

= RHS

$$(ii) a) \quad \frac{d^2x}{dt^2} = \frac{d(\frac{1}{2}v^2)}{dx} = -\frac{1}{2}e^{-x} \checkmark$$

$$\frac{1}{2}v^2 = -\frac{1}{2} \int e^x dx \Rightarrow \frac{1}{2}v^2 = +\frac{1}{2}e^x + k \quad (1)$$

$$\text{when } x=0, v=1 \Rightarrow k=0$$

$$\therefore v^2 = e^{-x} \quad (2)$$

$$\therefore v = e^{-\frac{1}{2}x} \checkmark$$

$$b) \text{ from } (1) \quad \frac{dx}{dt} = e^{-\frac{1}{2}x} \checkmark \Rightarrow t = \int e^{\frac{1}{2}x} dx$$

$$t = 2e^{\frac{1}{2}x} + c$$

$$t=0, x=0 \Rightarrow c = -2 \checkmark \quad (3)$$

$$\therefore 2e^{\frac{1}{2}x} = t+2 \Rightarrow e^{\frac{1}{2}x} = \frac{t}{2} + 1$$

$$\frac{1}{2}x = \ln\left(\frac{t}{2} + 1\right)$$

$$x = 2\ln\left(\frac{t}{2} + 1\right) \checkmark \quad (4)$$

You may ask for extra writing paper if you need more space to answer question 5

Question 6

BOS#:

a) $V^2 = 18 + 32x - 8x^2$
 $\frac{1}{2}V^2 = -4x^2 + 16x + 9$
 differentiate wrt x

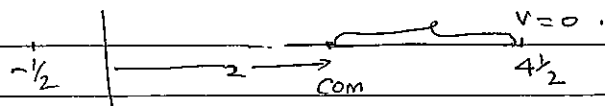
$\frac{d(\frac{1}{2}V^2)}{dx} = -8x + 16 \checkmark$
 $\frac{d^2x}{dt^2} = -8(x-2) \checkmark \quad \because \frac{d^2x}{dt^2} = \frac{d(KV^2)}{dx}$
 $\ddot{x} = -\eta^2 x \quad \text{when } x = x-2, \text{ and } \eta = 2\sqrt{2}$

$\therefore$ The particle is in SHM.

b) centre of motion is $x=0$, $\ddot{x} = -x = 2 \checkmark$
 period $= \frac{2\pi}{\eta} = \frac{2\pi}{2\sqrt{2}} = \frac{\pi}{\sqrt{2}} \text{ s} \checkmark$

c) Amplitude is the max displacement from the centre of motion. At this point $V=0$.

$4x^2 - 16x - 9 = 0 \checkmark$
 $4x^2 - 18x + 2x - 9 = 0$
 $2x(2x-9) + 1(2x-9) = 0 \Rightarrow x = 4\frac{1}{2}; -\frac{1}{2}$



amplitude $= 4\frac{1}{2} - 2$
 $= 2\frac{1}{2} \text{ cm} \checkmark$

You may ask for extra writing paper if you need more space to answer question 6

Question 7

BOS#:

a) $\frac{dp}{dt} = k\sqrt{p-1}$ when $t=0$ $P=10$
 $t=10$ $P=5$

$\int \frac{dp}{\sqrt{p-1}} = k \int dt$

$kt = \int (p-1)^{\frac{1}{2}} dp \checkmark$

$kt = 2(p-1)^{\frac{1}{2}} + C$

$0 = 2 \times 3 + C \Rightarrow C = -6 \checkmark$

$kt = 2(p-1)^{\frac{1}{2}} - 6$

when $t=10$ $P=5$ $10k = 2 \times 2 - 6$

$k = -\frac{1}{5} \checkmark$

b) $-\frac{1}{5}t = 2\sqrt{p-1} - 6$

when $t=20$

$-\frac{1}{5} \times 20 = 2\sqrt{p-1} - 6 \checkmark$

$2 = 2\sqrt{p-1}$

$p-1 = 1$

$p = 2 \checkmark$

$\therefore$ after 20 s pressure in the can is 2 Atmospheres.

Question 8

BOS#:

$$i) \quad x = vt \cos \alpha \quad \text{--- (1)} \quad y = vt \sin \alpha - \frac{1}{2}gt^2 \quad \text{--- (2)}$$

$$\dot{x} = v \cos \alpha \quad \dot{y} = v \sin \alpha - gt$$

at the highest point $\dot{y} = 0$. i.e.

$$\therefore v \sin \alpha - gt = 0$$

$$t_{\max} = \frac{v \sin \alpha}{g} \checkmark$$

In this time height reached is y .

$$\therefore y_{\max} = v \sin \alpha \left(\frac{v \sin \alpha}{g} \right) - \frac{1}{2}g \left(\frac{v^2 \sin^2 \alpha}{g^2} \right) \checkmark$$

$$= \frac{v^2 \sin^2 \alpha}{g} - \frac{v^2 \sin^2 \alpha}{2g}$$

$$= \frac{v^2 \sin^2 \alpha}{2g} \quad \text{--- (A)}$$

$$ii) \quad \text{from (1)} \quad t = \frac{x}{v \cos \alpha}; \text{ sub this in (2)}$$

$$y = v \sin \alpha \cdot \frac{x}{v \cos \alpha} - \frac{1}{2}g \frac{x^2}{v^2 \cos^2 \alpha}$$

$$y = x \tan \alpha - \frac{1}{2}g \frac{x^2}{v^2 \cos^2 \alpha} \quad \text{--- (3)} \checkmark$$

when the ball ~~pass~~ clears the wall $x=d$, $y=h$

Sub in (3)

$$h = d \tan \alpha - \frac{1}{2}g \frac{d^2}{v^2 \cos^2 \alpha}$$

$$\Rightarrow \frac{v^2}{g} = \frac{d^2}{(d \tan \alpha - h) 2 \cos^2 \alpha} \checkmark$$

$$\text{Sub } \frac{v^2}{g} \text{ in } (A')$$

$$y_{\max} = \frac{\sin^2 \alpha \cdot d^2}{2(d \tan \alpha - h)} 2 \cos^2 \alpha \quad \checkmark$$

$$= \frac{d^2 \tan^2 \alpha}{4(d \tan \alpha - h)}$$

$$= \frac{1}{4} \left[\frac{d^2 \tan^2 \alpha}{d \tan \alpha - h} \right] :$$

You may ask for extra writing paper if you need more space to answer question 8